UCLA College | Humanities English

DEPARTMENT OF ENGLISH
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February 3, 2026

Subject: Letter of Recommendation for Narayan Gopinathan

Dear Colleagues:

I am delighted or rather honored to write a letter of recommendation on behalf of Narayan Gopinathan, who was one my TAs for the Fall 2025 Food Cluster. Let me begin by saying that in my twenty years as a teacher, mentor, and research scholar at three different universities (Fordham, Yale, and UCLA), I have not met a single TA or, for that matter, a single graduate student as attentive, knowledgeable, proactive, and efficient as Narayan. Words like these may seem formulaic or empty but let me offer a few examples of how Narayan has inspired not just the students in his two sections but also me and, in so doing, remains, in my judgment, the best of the TAs I have had so far.

To begin with, Narayan went above and beyond the role of a TA insofar as he answered the learning needs of students in both his sections. On many occasions in the classroom, I observed Narayan interact with students and found him to be both deeply attentive to the questions posed by them and deeply knowledgeable in his responses. When a student approached Narayan about any aspect of the course, he did not assume what they were saying. Rather, he paid close attention to their words and then, like a latter-day Socrates, engaged in a dialogue in ways that enriched not only the student in question but also any others that happened to be listening in (as many of these dialogues took place in public). For instance, when a student asked about the Green Revolution, Narayan did more than offer a direct answer to a single student. Rather, he first asked other students for their responses and, building on them, went on to provide contexts for the changes in our food systems over five decades, and thereby linked the original question to the central topic of the course. In redirecting a question from a student to other students, Narayan empowered the class to think together as a community and only then added his inputs. That is, Narayan demonstrated that learning does not proceed vertically (from instructor to student) but horizontally (from student to student) and that by redirecting questions the class can learn actively and collectively. Witnessing Narayan teach students by having them think for themselves was for me a most inspiring lesson.

Allow me to describe another thing that Narayan did that I thought was particularly beneficial for our students' learning. Given that my lectures dealt with animals in food systems across time, students often had trouble understanding why a food system evolved as it did. What Narayan accomplished in the discussion sessions was to provide the larger contexts of the readings and the lectures. That is, he went out of his way to introduce primary (and often canonical) sources that students could use to revisit the topics I had lectured on. For example, after I had introduced students to Plato's and Aristotle's views on food and animals, Narayan introduced us to a pupil of Aristotle, Theophrastus, whose writings on the abstention from meat students and I found to be a valuable contextual framework for plant-based diets in classical antiquity.

Likewise, at one point in a lecture, I indicated that language helps distance ourselves from the animals we eat, noting that in English we use one set of words for the meat of mammals and another set for the mammals themselves. In a discussion session that followed, Narayan offered instances of such English usage but went further by asking students if this was so in other languages. In raising this question in what was a linguistically diverse class, Narayan motivated multilingual students to provide answers and serve as active learners. In other words, Narayan was not himself clarifying what I had taught but having them contextualize it and thereby ensuring that they were approaching the lectures and readings with renewed perception and intensified enjoyment. After what was a lively discussion of how different languages distinguish between animals and their meat, Narayan pointed out that Chinese has the character 肉 that shows ribs and thereby makes vivid the connection between the sign and the animal indicated by it. In offering such a striking example, he did not simply review what I covered in the lecture but complemented it and thereby captured and sustained the interest of the entire class.

Meeting students where they are, Narayan inspires and enables them to participate as agents in the adventure of ideas not just within the classroom but also outside it. At one of several food-related events held at UCLA that Narayan invited students to attend, I spoke at length with a former student of his – a sophomore who attributed her success in her academic career to him alone and highlighted his continued mentorship of students long after their courses ended.

A model of tireless industry and profound generosity, Narayan is the rare TA that both instructors and students would be fortunate to work with. He is patient, cooperative, thoughtful, and above all a passionate and caring scholar-teacher. A delightfully vibrant presence wherever he happens to be, he stands out as the best graduate TA I have ever had. I highly recommend Narayan without any reservations for the Distinguished Teaching Award. Please feel free to contact me should you have any questions.

Arvind Thomas, PhD.
Associate Professor of English

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Self-statement – Narayan Gopinathan

I have completed ten quarters as a teaching assistant or teaching fellow at UCLA, having taught classes as varied as life cycle assessment, water management, big environmental data, statistics, econometrics, and tropical ecology. I have secured stellar reviews each time, and made a difference in my students' lives. Each time I hold a discussion section, I try to not just explain course material, but go beyond that by providing context for it. For example, two years ago, I was the TA for a statistics class for students doing their MPP. The students were a diverse group, as many had not taken math classes since high school, and some were older than me. For that reason, I had to make it relevant for them by giving examples of how these concepts applied to the real world.

I have also supported them by being there for them, as someone who can listen to them. After a tough midterm in the statistics class, one student felt so discouraged. After I listened to her vent about the course, she told me, “I was going to drop out, but you know what? Because you’re here, I’ll stay.” I served as an interface between the students and professors, to make the exams fair.

In the past two years, I have been a teaching fellow in the food cluster, a series of courses that teaches freshmen about the impacts of food systems, particularly the environmental and ethical impacts of animal agriculture. In this capacity, I have had the opportunity to teach something that I am passionate about. I have continued to support my students by providing context about course material. For example, in discussions of how the global food system evolved, I explained how the world was very different fifty years ago, as global population was lower but there was real concern about food shortages. To avert these crises, scientists developed new ways of growing crops that produced more yields, but these came at an environmental cost because they required non-renewable inputs such as fertilizers, and pesticides. Similarly, when discussing the reliance of American agriculture on migrant labor, I told them about the history of the US-Mexico border. When discussing animal rights, I showed them a picture of me with my dog, describing his personality, and asked whether other animals might also have a personality. I showed this picture along with a spider I rescued from my bathtub before showering, and their eyes widened as they considered the different way of viewing spiders, which many people would kill on sight.

One of the best things about teaching has been empowering my students to realize their full potential. For example, in sharing sources of data with one of my students, I helped her complete an amazing project to propose turning vacant lots in LA into gardens. I sent her information about a conference where she could present the work, and she won the conference’s top award. She also won the Undergraduate Library Prize and the Tom Davey

Science Writing Award here at UCLA. She did not believe in herself, back in high school, but credited me with helping her believe in herself.

Even as I provide a boost to my top achievers, I also make sure to help the students who struggle a bit. One of my students last year produced a first draft of an essay that was not cohesive. It made many good points but did not cohesively come together as an essay, and many sentences were poorly written. I made many specific comments on the essay, and gave a low score for the rough draft, given that many required rubric items were absent. He felt discouraged, and unsure where to start. I told him "I know you can do it." I told him that, though it is hard for both him and for me, it is my responsibility to improve his writing. I drew a comparison to my own history, when I took a freshman writing seminar in college, which was tough but improved my writing. I gave him a direct writing tip to think about his elevator pitch, to encapsulate his essay in a nutshell, to help him clarify what he is trying to say. After that, he focused his ideas and wrote a great essay.

Supporting the students in their learning journeys, and enabling them to achieve their potential, is among the best things I have achieved in graduate school.

Statement of Teaching – Narayan Gopinathan

There are two essential elements of teaching. First, it is important to explain the course material in a way that is accessible to them. I go beyond what the professors teach in the classroom, providing context for course content. Second, it is important to show that you care about the students. I do that by listening to them, and making space for their own voices, in regard to course content or anything else. In grading, in writing essays, and in giving presentations to them, care for the students and their experiences is the key to a good learning experience. A great teacher isn't someone who just delivers great lectures, but builds community with their students.

I strive to create an engaging and inclusive learning environment, by caring about the students and their perspectives, and being mindful of what diverse perspectives they may have. For example, once a professor for an unrelated class described principles of reproductive justice. I advised him to avoid pushing his own views on the topic, for if one student holds views that run counter to his, they might feel alienated. This mentality of consideration creates a safe space for students to speak up and share their perspectives. I engage with what they share, to build upon it, and find connections between their thoughts and experiences and the bigger ideas we engage with in the course. This way, students feel safe, empowered, and motivated to share their own thoughts and engage with the course content, and with the community in the classroom, without fear of penalty or disapproval. I draw connections to concepts and examples that the students may be familiar with. One way I do that is by relating them to the places where they are from.

In the food cluster this quarter, we had an assignment to build a GIS storymap – essentially a website with maps and geographic data. Though the main thrust of the course is to grapple with the issues caused by animal agriculture, one student asked if she could write about the land use change in her hometown of Riverside, California, where agricultural land, such as citrus orchards, has been lost to industrialization. I discussed the issue with her, about how so much agricultural land has been lost to industrial warehouses in the Inland Empire, and encouraged her to write about that issue. I found some data on the issue and shared it with her, to help her strengthen her research.

This applies to communications outside of the classroom as well. For example, as part of my community outreach, I attended a science fair at a high school in Bell Gardens, California, which is located near a railyard. I told students about my interest in reducing emissions from the locomotives, to improve air quality in communities like theirs. The common thread between my communications with students inside and outside the classroom is making the content relevant to students' lives.

I also subtly remind them that things could be different elsewhere. When discussing national average public health data, I use phrases like “here in America,” to ground the data in this country while also reminding them that America is not the only country, and that in other parts of the world, the data and situations can be different. As much as I draw connections to the students’ lives, I also bring examples of situations in other places, to expand their viewpoints of the world outside their own bubble. For example, in my class on food, transportation, and environment, I told the students about the controversial proposal to build a road through Izembek National Wildlife Refuge in a remote part of Alaska, which would improve life for the people in the region, but devastate a local ecosystem, to illustrate the tradeoffs between development and environmental protection. This encouraged them to think about ways that we can address these issues, in ways that protect the ecosystem and enable human well-being.

In my discussion sections, I make sure to give full context for the course content. When offering examples of problematic policies, I explain why they were instituted, to provide context and clarity. For example, I explained how when the US biofuel mandate was passed at a time when electric vehicles were not available, and so biofuels were seen as the best option to reduce oil use in cars, and the problems with this policy, which induced land use change, became more apparent later. One bright student, who was more conservative than most at UCLA, said this shows how problematic policies are not designed with malice. Later in that discussion, another gave counterexamples of problematic policies that may have malicious intent. This fostered a lively discussion about public policies and their impacts, and how the best policies can be designed to address environmental and economic problems.

Throughout my teaching career, I have not viewed myself as superior or my students as less than me or below me. I have instead viewed them as students on the same path as me, with myself just a few steps ahead of them, as I try to bring them along with me, with an expanded understanding of the world and of the course content. This moves away from academic hierarchy, fostering a dialog-based conversational environment that enhances the learning environment and provides a safe space for students to find and share their own voices.

I find that the most important aspects of teaching are connecting the material to students' lives, while also providing examples of how they can be different in other circumstances. In the food cluster, as well as in other courses, I peel back layers of reality, to explain why things are the way they are. This builds community with students, and expands the students’ worldview.

Spring seminar: Transportation, food, and environment. Spring 2025

Location: Public Affairs 1343. Time: Wednesdays from 9 am to 12:00 noon.

This course is about the nexus between transportation, food, and the environment. Biofuels are one major component of this nexus, as they can cause the diversion of food crops to fuel tanks, but there are other ways that these topics link. For example, construction of roads can lead to agricultural development in rainforest areas and induce deforestation; while zoning laws that practically mandate car use can induce urban sprawl and thus lead to the loss of agricultural farmland. Roads are a key component of agricultural development, because they enable farmers to move their crops to the market, and thus enable people to live in a different place from food production. Diesel fuel is used to refrigerate trucks for frozen food, and the refrigerant chemicals cause warming. The nexus is more interesting and complex than you might think.

We will have weekly readings that relate to this nexus, along with climate change and its solutions more broadly. We will foster a Socratic seminar style discussion each class session, based on the readings for the week. I will also give you time to work on writing assignments in class.

Learning outcomes include:

- Analyzing tradeoffs, and assessing benefits and harms, associated with the use of biofuels, and all other environmental choices more broadly.
- Gaining analytical skills related to consequential and attributional life cycle assessment and policy analysis.
- Gaining writing skills related to the composition of policy memos, op-eds, and scholarly articles about the tradeoffs related to biofuels.
- Making connections between food, environmental, and transportation policy.
- Be aware of and empathize with different perspectives on these choices.
- Independently research biofuel and transportation policy and related topics.

It will develop foundational knowledge with presentation, lecture, and large and small group discussions. It will develop an integrated curriculum which will help them develop a detailed and integrated understanding of the impacts of food and energy systems through concept mapping, the development of cross-disciplinary cases, and providing multiple examples within and across contexts.

Instructor: Narayan Gopinathan. Email: ngopinathan@g.ucla.edu
I am here to support you, so feel free to reach out anytime.
Office hours: Thursdays 9:30 to 10:30 am in Life Sciences 2306. You can also reach out to me to schedule another time to talk by Zoom or in person.

Course materials: Bring a laptop, pen, and paper to class every week.
Textbook: “Own your perspective” by Tamar Christensen and Randall Fallows. Available online for just $15.95 at the following link: https://students.flatworldknowledge.com/course/2612582

Learning goals:

Foundational Knowledge:

- Compare and contrast the advantages and disadvantages of biofuel policies.
- Describe the differences between impacts of biofuels and other options for transportation fuels.
- Associate the food policies with transportation policies and vice versa.

Application

- **Critical thinking -** Analyze the impacts that biofuels have on the environment.
- **Practical thinking** - Determine the best options for policies with respect to biofuels.
- **Creative thinking** - Compose different types of writing - op-ed, scholarly article, free-response to reading, etc.
- **Managing complex projects** - Strategize the best ways to develop policies related to food, fuel, and the environment.
- Identify sources and resources for energy.

Integration

- Associate the impacts of transportation choices on food systems and food choices on transportation systems.
- Connect the choices in one country with impacts on another.
- Integrate different perspectives on the issues related to biofuels, agriculture, and the environment.

Human Dimension

- Be aware of the impacts of their own choices with respect to food and transportation on the global environment and on global hunger.
- Empathize with the global poor who are hungry, and with different perspectives on issues related to biofuels.
- Feel confident in developing their own perspectives on their issues, while being respectful and empathetic towards other perspectives.

Caring

- Developing an environmental ethic to care about the impacts of their actions on the nonhuman world, and on other people.
- Express their own opinions on issues related to biofuels, food policy, and the environment.
- Value the intact natural habitat and biodiversity.

Learning how to learn

- How to be a better learner, engage in self-regulated or deeper learning.
- Frame useful questions - what are the right questions to ask about the impacts of policies?
- Identify sources and resources for knowledge.
- How to inquire and construct knowledge - how to develop the right questions and find the answers on your own.

Writing assignment:

The main writing assignment for the class is to write a research paper on a topic of your choice related to this nexus of food, environment and transportation. It can be about biofuels or something else related to this nexus. Get creative - as long as you can explain why it is related to these topics, you can write a great paper. The paper should be 6 to 8 pages long, double spaced, or 1500 to 2000 words. The word count does not include references. You should have eight to ten references, and they should be quality sources. At least four of them should be scholarly sources such as peer-reviewed journal article, and at least one source should be a book.

You will also write an op-ed which makes the same points as your scholarly paper, but formatted differently for a different audience. The idea is that it would be published in a newspaper such as *The Conversation*. This should be 2-3 pages long, or 500 to 750 words.

To get full credit, the writing should go beyond summarizing the content of the sources. I ask you to come up with your own thoughts, ideas, which are grounded in evidence. This is not a persuasive essay. You are not being asked to make one simple argument (i.e. “biofuels are good” or “biofuels are bad”) and hammer it home. If you do that, you will not get full credit. You are being asked to explore the nuances of whatever issue you choose to explore, and consider the pros and cons of any argument you make.

AI Policy:
Students are required to write essays on their own, to come up with their own thoughts and ideas. They are encouraged to avoid the use of AI for writing, and do all the writing on their own. Generative AI programs can be used as a learning tool, but not as a replacement for a student’s own thoughts and ideas. If they choose to use AI, students must engage critically with AI outputs rather than accepting them uncritically, because AI can be wrong just like anything else can be. It is not acceptable for students to have ChatGPT or other generative AI tools write essays for them.

Course schedule

Each week, a deliverable related to the writing assignment will be due on BruinLearn the Saturday after the week, at 11:59pm.

Each week, there are assigned readings. It is expected that students have read the assignments before coming to class on Wednesday. For many assigned readings, you are assigned a portion of an article or other document, but encouraged to read the whole thing!

Each week, there will be 2-3 readings assigned, for a total of 20 readings throughout the class, which correspond to 20 students. Each class session, 2 or 3 students will be assigned to lead a discussion of the readings. Before each class session, every student has to write 1 or 2 questions for discussion and submit them on BruinLearn. We will discuss these questions in class as part of the Socratic seminar. You are expected to actively participate in each discussion, rather than staying silent - the class won't work if you don't talk. Participation will count for points. To earn the participation points, you just have to make at least one question or comment related to course material each day in class. Each class we will do a free write related to the readings or other course material. It should be written on paper – laptops and phones should be turned off. If you have to miss class, send me a reading response by email.

Instructions for facilitating Discussion:
One major assignment for the spring seminar is to facilitate a discussion on a reading. To do a good job of this and get full credit, you have to read the piece carefully and think of main points that you want to share with the class. However, this is not a presentation. It is a way to start a discussion.
Think of some questions that relate to the readings and enable the class to dive into a deeper discussion analyzing it. Take a critical perspective. What in the readings surprised you? Was there anything counterintuitive? Was there anything you disagreed with?

Week 1: **Discuss climate change and its broader context.**
Readings on content:
Reading A: Paris Agreement Preamble and articles 1-4. ADOPTION OF THE PARIS AGREEMENT - Paris Agreement text English

Optional podcast: Outrage and Optimism, "How big oil is holding back progress" first 27 minutes of the podcast. How big oil is holding back progress on the Paris Climate Agreement
Reading B: IPCC Summary for policymakers Sections B and C. IPCC_AR6_SYR_SPM.pdf

In class: After discussing readings, we will brainstorm topics for papers. We will put your ideas on the board and draw out the connections between them like a spiderweb diagram.

Assignment for Week 1: Think of a main topic for your paper, along with what you know about it coming into this class.

Week 2: Focus on developing researching and writing skills.
Reading A:
https://guides.law.stanford.edu/c.php?g=1255722&p=9200578
AND "How do I know if a source is credible"
https://hbl.gcc.libguides.com/research/credible

Reading B: Researching with Wikipedia - Wikipedia. -
https://en.wikipedia.org/wiki/Wikipedia:Researching_with_Wikipedia
AND LMU guide to researching with Wikipedia.
https://libguides.lmu.edu/c.php?g=323798&p=2169075

Reading C: Researching with AI - Charlotte Huff. The promise and perils of using AI for research and writing.
https://www.apa.org/topics/artificial-intelligence-machine-learning/ai-research-writing

AND Georgetown Library AI resources.
https://guides.library.georgetown.edu/ai/tools

Reading on analysis: *Own your Perspective,* Chapter 1-2: "Developing Assertions from a Close readings of Examples".

In class: We will go to the library to look for books that can serve as sources.

Assignment for Week 2: Find at least one book that you can use as a source. Submit the name and title of that book, and some notes about it.

Week 3: **Climate change solutions.**
Reading A: Drawdown 2020 report, Section 1.2 on Food.
Reading B: Drawdown 2020 Report, Section 1.4 on Transportation
Reading C: Drawdown 2020 Report, Section 2.1 on Land Use Sinks.

Reading on analysis: *Own Your Perspective*, Chapter 3: " Explanations, Significance, and Sources".

Assignment for Week 3: Thesis and annotated bibliography:
The thesis should be a paragraph describing a main point you are trying to make.
The annotated bibliography should have at least 6 sources (including the ones you found in weeks 2-3) and notes about what you found in each one, and their relevance to the topic.

Week 4: Biofuels for hard-to abate sectors such as marine shipping and aviation.
Reading A: Jeffrey Ampah, et al. Journal of Cleaner Production. Reviewing two decades of cleaner alternative marine fuels: Towards IMO's decarbonization of the maritime transport sector. Section 2 gives an overview of the options for Marine Decarb. The assignment is sections 1-2 of this, but as always, students are encouraged to read the whole thing.

https://www.sciencedirect.com/science/article/pii/S0959652621030675?via%3Dihub

Reading B: Ian Shine. World Economic Forum. What is sustainable aviation fuel and why are only 0.1% of flights powered by it?
https://www.weforum.org/stories/2023/11/what-is-sustainable-aviation-fuel/

AND Ben Purvis. The Conversation. There isn't enough 'sustainable' aviation fuel to make a dent in our emissions – and there won't be for years. https://theconversation.com/there-isnt-enough-sustainable-aviation-fuel-to-make-a-dent-in-our-emissions-and-there-wont-be-for-years-249270

Reading on analysis: Own Your Perspective, Chapter 4: The Traditional Essay.

Assignment for Week 4: Develop an outline for the whole paper. It can be in bullet point form.

Week 5: Biofuels impacts on environment.
Reading A: Biofuels: An obstacle to real climate solutions. Report by Transportation & Environment and Oxfam.
Reading B: Dan Lashof. WRI. EPA's New Renewable fuel standard will increase carbon emissions - not lower them. https://www.wri.org/insights/us-renewable-fuel-standards-emissions-impact

Assignment for week 5: Write a first draft of an op-ed based on the same topic you are researching. The style should be similar to what we read in *the Conversation*.

Week 6: Land use and land use change.
Reading A: Searchinger et al. Use of U.S. Croplands for Biofuels Increases Greenhouse Gases Through Emissions from Land-Use Change. Science. 2008.
https://www.science.org/doi/10.1126/science.1151861
Reading B: Introduction and conclusion (Sections 1 and 4) of the following paper: Bacor and Prus. Infrastructure development and its influence on agricultural land and regional sustainable development. Ecological Informatics.
https://www.sciencedirect.com/science/article/pii/S1574954117302352#s0080

Assignment for Week 6: Edit and revise the op-ed, and submit the revised op-ed.

Week 7: Impacts of roads:
Reading A: Sections 1-3 of the following paper: Yogita Shamdasani – Rural road development & Agricultural production: The evidence from India.
https://www.sciencedirect.com/science/article/pii/S0304387821000638

Reading B: Eric Wagner. 6PPD: The secret killer of Puget Sound salmon.
https://www.nature.org/en-us/about-us/where-we-work/united-states/washington/stories-in-washington/6ppd-secret-killer-of-puget-sound-salmon/
AND Mark Johansson. How the Interoceanic Highway ushered in a new era of deforestation and social upheaval in the Amazon.
https://www.worldwildlife.org/stories/how-the-interoceanic-highway-ushered-in-a-new-era-of-deforestation-and-social-upheaval-in-the-amazon

Assignment for week 7: Write 3 pages of content, which should consist of primarily body paragraphs.

Week 8: Transportation of food
Reading A: Hannah Ritchie. You want to reduce the impact of your food? Focus on what you eat, not whether your food is local. January 24, 2020. https://ourworldindata.org/food-choice-vs-eating-local

Reading B: Ayesha Tandon. Food miles have a bigger climate impact than thought. Carbon Brief. 2022. https://www.carbonbrief.org/food-miles-have-larger-climate-impact-than-thought-study-suggests/

Week 8: Write draft for 6 pages for final report, not including introduction and conclusion.

Week 9: Cold Chain:
Reading A: Introduction to “Frostbite” by Nicola Twilley.

Reading B: The Kigali Amendment: Congress’s First Approved Climate Treaty in Decades | Georgetown Environmental Law Review | Georgetown Law

In class: First 5 presentations. Students will give a presentation about their main topic.

Week 9: Write a draft of the entirety of the report and submit it.

Week 10: Presentations:

In class: Remaining 15 presentations. They can target 6-8 minutes each, with time for questions. There will be 12 minutes available for each presentation.

Week 10: Write the final draft of the article and submit it.

Course Attendance:
You are expected to show up on time to every class, and will be graded on attendance and participation. To earn full credit for participation each day, you have to make at least one comment or question related to course material or readings during the class. The Socratic seminar won't work if everybody is silent. You can miss 1 class without losing points, but you are expected to email me if you will be absent for any reason. Since there are only ten classes over the quarter, missing even one is a big deal, and two is 20% of the course.

At the beginning of class each week, there will be a reading quiz. The reading quiz will be a simple response to the readings for the week, written on pen and paper, with laptops and phones turned off, to ensure everybody did the reading. Anybody who is absent from class should submit a reading response by email.

Grading
The whole course will be out of 600 points. Since each point will be weighted equally, this will make the grading structure easy to understand.

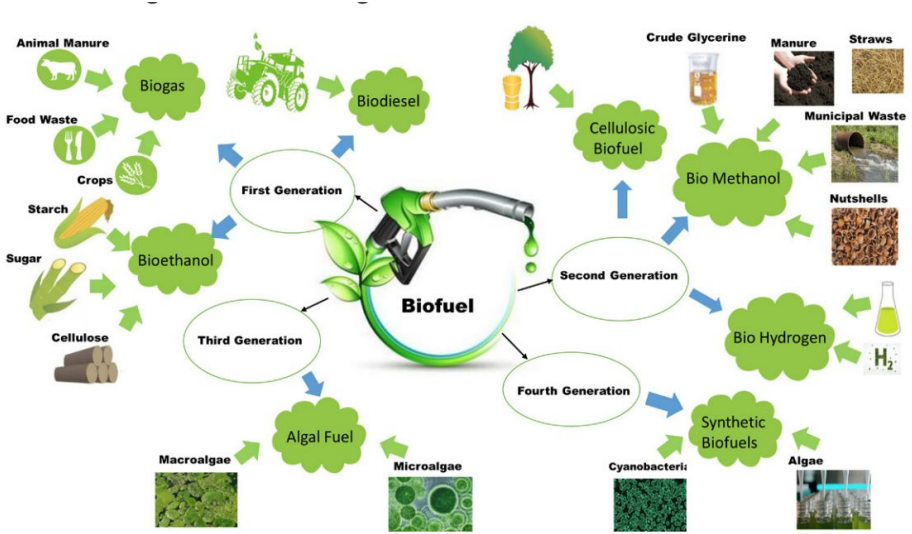
	Points / assignment	Assignments	Points Total
Posting discussion questions on BL	10	9	90
Reading quizzes in class	10	9	90
Attendance and participation (each class)	20	9	180
Facilitating discussion on reading	20	1	20
Components of course paper	20	10	200
Presentation	20	1	20
Total			600
Threshold for A-			540

Instructor Information: Narayan Gopinathan. My email: ngopinathan@g.ucla.edu

Food Cluster Discussion
Winter week 5

My spring seminar: Transportation, food and environment

- We cannot give PTE numbers as of now.
- Biofuels will be only one focus of the seminar.
- More broadly about the nexus between food, transportation, and environment.
- Can you think of other connections between food, transportation, and environment?



Transportation, food, and environment

- Zoning - car-oriented development can eat up farmland. Orange County was once orange groves, and now is suburban sprawl.
- Road Building in rural areas enables agricultural development, which can also cause deforestation.
- Diesel is used in refrigerated trailers for cold chain of frozen foods.
- Salmon used for food by indigenous people are impacted by road culverts and tire particulates.









Animal ethics

- What did we think of the readings from last week?
- The way we use language of meat hides the individuality of animals.
- Do we call an animal it, he, or she? Something or someone?

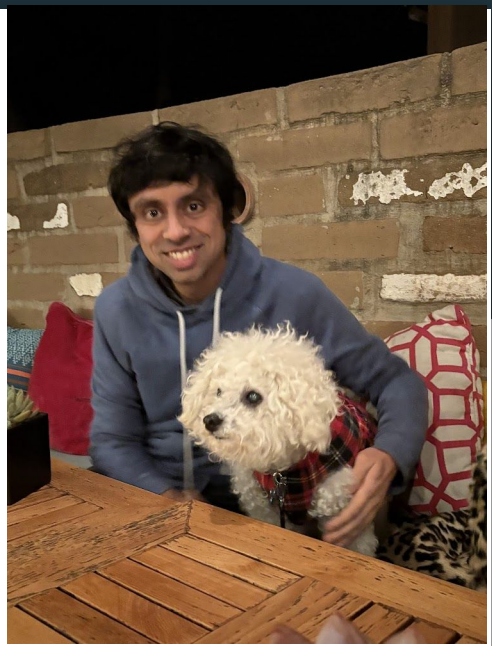






Animal ethics

- Animals have personalities.
- Many of the animals that people eat are mammals and birds
  - They share our biology.
  - They bleed red and feel pain like we do.
- We can care about creatures that have different biology too.
- Break into smaller groups to discuss these topics.

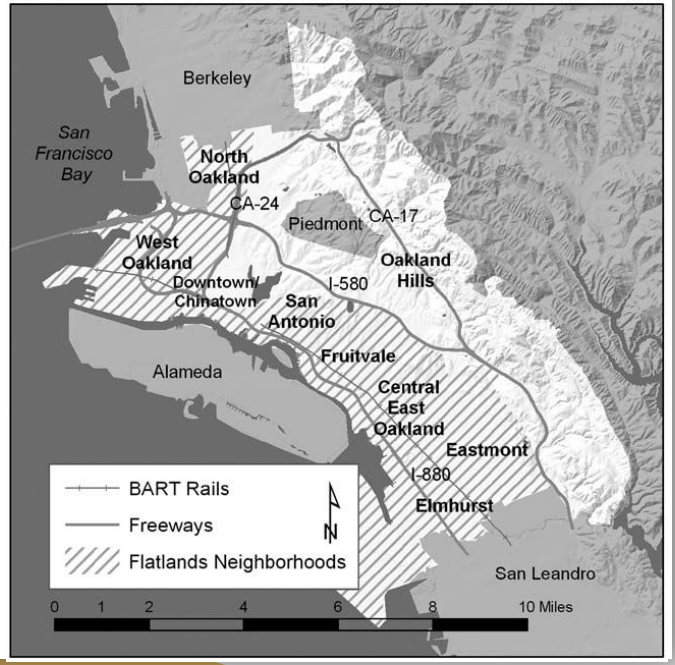






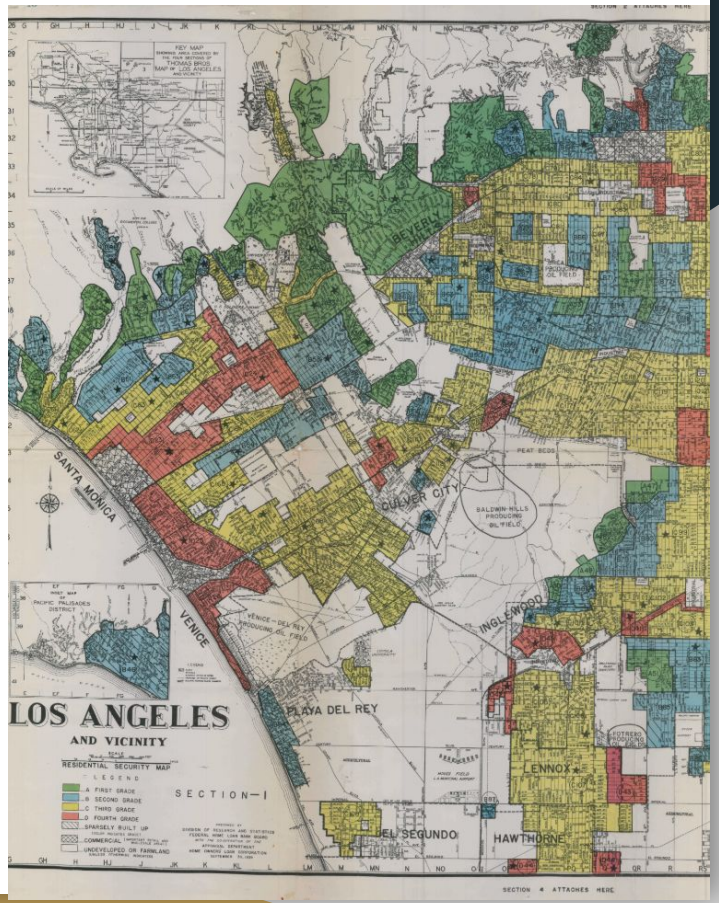
Redlining and food deserts

- In many American cities, certain neighborhoods were designated as “hazardous” for lending, where people of color lived.
- These are still the most underprivileged neighborhoods.
- Access to fresh healthy food there is more limited.
- In Oakland, trucks can’t use I-580 through the hills, so they use I-880 through the flatlands, polluting the flatlands air with diesel.



Structural racism in LA

- Westwood had a racial covenant which prohibited people of color from living here.
- Harold and Edwin Janss - namesakes of Janss Steps - sold the land for campus to the UC to build UCLA.
- They developed Westwood Village with a restrictive racial covenant.
- Interstate 10 bulldozed a Black neighborhood called Sugar Hill.



Work on scholarly paragraph and writing assignment

- Take the post-lecture paragraph you wrote last week, and update it with scholarly sources.
- Ideally have at least two sources.
- You don’t have to answer the questions about how to categorize the solution.
- Describe the solution you have and its impacts, with clear references that describe both the solution and the broader issue it addresses.
- Select topic (problem and solution) for final assignment, including identifying target audience. List and properly cite at least 5 sources related to your chosen topic and provide an annotated bibliography.

Transportation, Food and Environment

Week 3

Reading quiz

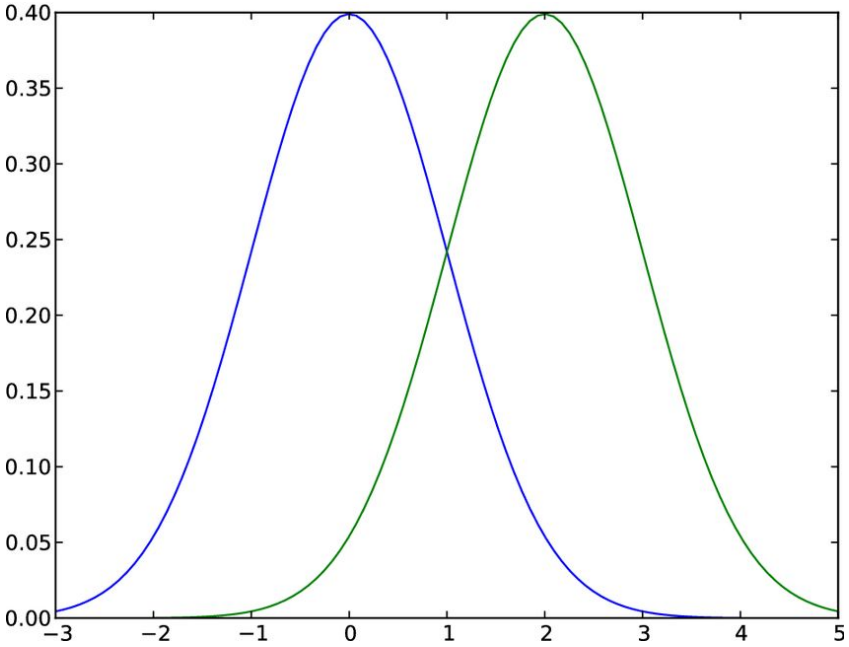
- What were your main takeaways from the Drawdown Report?
- Specifically, what did you learn from the sections on
  - Food
  - Transportation
  - Land use
  - Any other section you read?
- For full credit, mention all three of food, transportation, and land use.

Temperature targets

- The Paris Agreement talks about targets like 1.5°C and 2°C above the preindustrial baseline.
- This means the average temperature before fossil fuels impacted the climate.
- The period from 1850 to 1900 is generally considered a good baseline, and the first period for which global temperature data is available. Global average temperature then was about 13.5°C or 56.3° F.
- This means that we want to keep global average within 2°C of that baseline.
- Source: Jennifer Chu. MIT News. August 27, 2023. Explained: The 1.5 C climate benchmark | MIT News | Massachusetts Institute of Technology

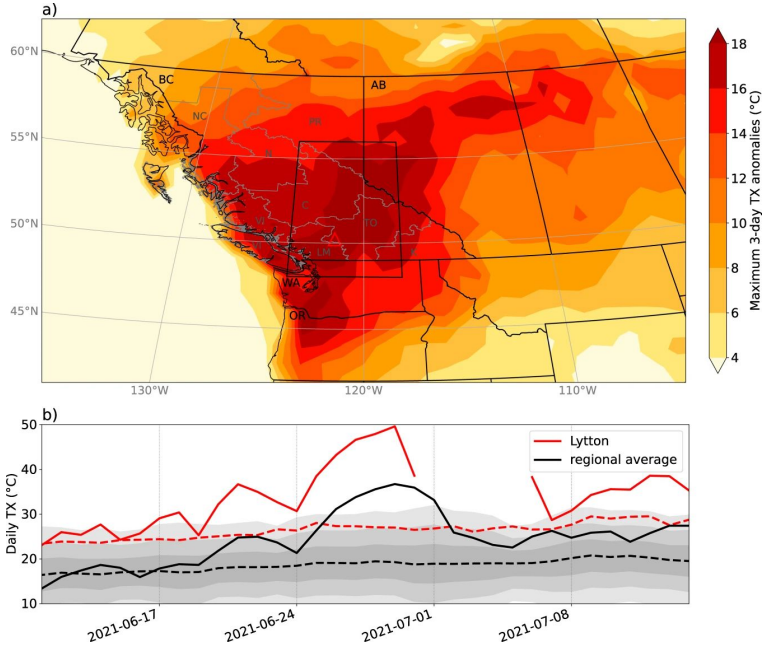
What does it mean to warm 2°C?

- Temperature can vary up to 18°F or 10°C between night and day in one place - why does warming of 2° matter?
- 1°C is 1% of the way from the freezing point to the boiling point.
- Warming 2°C means the whole planet warms 2% of the way from freezing to boiling.
- When the global average shifts that much, the extremes shift much more.



Extreme weather events

- June 29, 2021, temperature in Lytton, BC, in Canada, hit 49.6°C or 121.3°F.
- When the distribution shifts, events that were previously impossible start to happen.
- Rachel White, et al. The Unprecedented Pacific Northwest heatwave of June 2021. Nature Communications. 2023.



Drawdown - Refrigeration

- Refrigeration is also important for food systems and for emissions.
- Refrigerant chemicals are used as fluids for pumping heat in fridges and air conditioners.
- Refrigerant has to have the right properties: must become a gas after absorbing heat in the fridge, then become liquid when compressed, to release heat outside the fridge.
- Ammonia is toxic if it leaks, so Thomas Midgley invented CFC.
- CFC hurt ozone layer so then fridges transitioned to HFC under Montreal Protocol.
- HFC has a high GWP, so Kigali Amendment phases that out too.
- Now they are transitioning away from that to other chemicals.
- Refrigerant management is important lever we can pull under Drawdown.

Drawdown - food emissions

Vishnu Potharaju can facilitate discussion for solutions to food emissions.

Drawdown - transportation emissions

Alex Kolodziej can facilitate discussion on solutions to transportation emissions.

Drawdown - Land use sinks

Aneil Sandhu can facilitate discussion on land use sinks.

Ten minute break

Own your perspective

- Reviewing key details - Metaphors and analogies
- Developing worthwhile assertions - must be defensible, rooted in evidence.
    - Statements of fact, classification, taste, opinion, intention - not main analysis.
- Worthwhile assertions reveal a perspective that affords further exploration.
    - So what? Why does this matter? Most important question to answer.
- Provide background information.
- Considering other sources:
    - When was it published?
    - What is the author’s expertise? What are their biases?

AI

- Stand on the shoulders of giants?
- Ideally, independent thought goes beyond capability of AI.
- ChatGPT and other language models can do so many things - a educated person can do more.

Work on essay

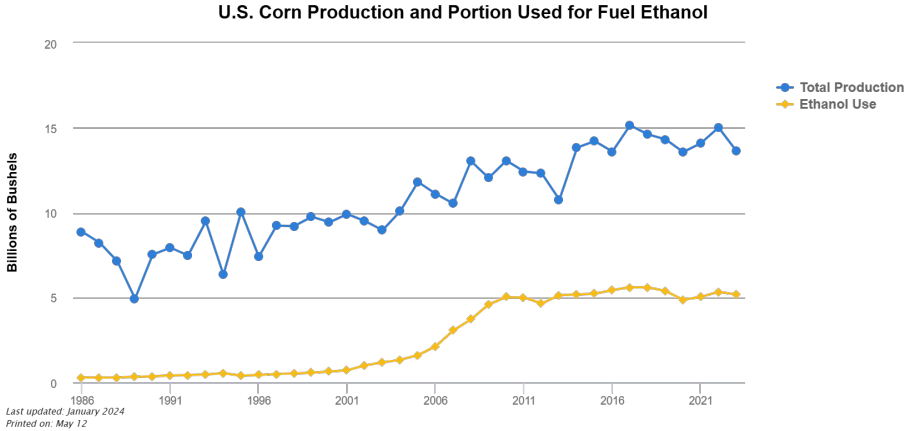
- By Saturday, have a thesis - a worthwhile assertion - and six sources.
- I will discuss your essays with each of you.
- Please let me know if you feel like you are struggling or need extra help.

Food, transportation, and environment

Spring 2025, Week 7

Last bit on biofuels

- Ethanol based biofuels consume almost 40% of US corn production.
- This contributes to overproduction and environmental damage.
- Transportation electrification and public transit can fix this.
- Source: USDOE Alternative Fuels Data Center. Maps and Data - U.S. Corn Production and Portion Used for Fuel Ethanol. https://afdc.energy.gov/data/10339



Trade policies and shipping emissions

- Tariffs are reducing demand for shipping, and reducing oil prices enough to hurt oil industry.
- For weeks, transpacific shipping was all but stopped because of tariffs.
- A recent deal paused tariffs, so shipping is resuming.
- Source: The Economist. Donald Trump is throttling America's oil industry. May 11, 2025 https://www.economist.com/business/2025/05/11/donald-trump-is-throttling-americas-oil-industry
- Scott Horsley. First, relief. Then, a scramble to get stuff in after U.S. cuts China tariffs for now. NPR. https://www.npr.org/2025/05/13/nx-s1-5395866/us-china-tariffs-trade-deal



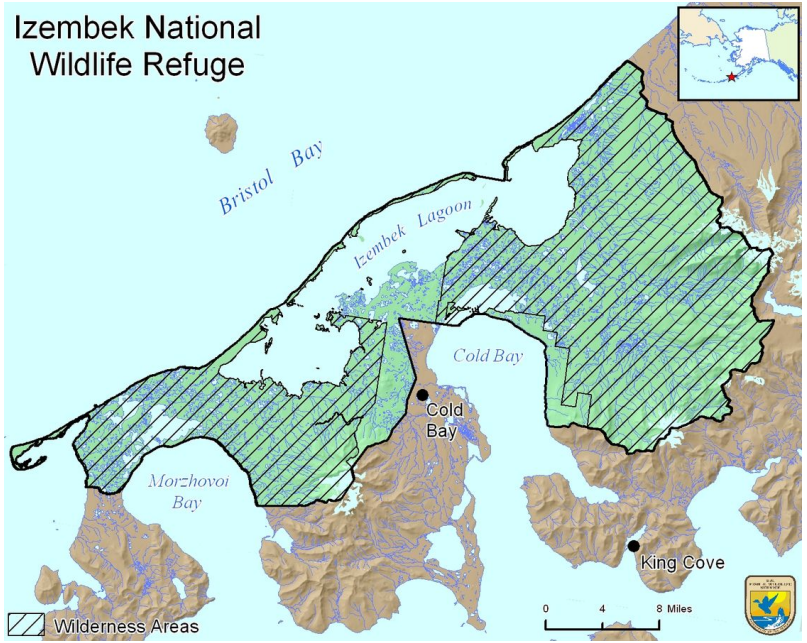
Road, Rail and Wildlife

- Roads have a big impact on wildlife because there are so many of them, and so many vehicles travel on them.
- Railways typically have less impacts, but they do too.
- HS2, under construction in UK, is planned with a tunnel to protect endangered bats from getting hit by trains.
- Bechstein's bats are among the world's rarest bats.
- This costs £100 million, equivalent to $133 million.
- UK politicians are now saying they should build without the tunnel to save cost and time.
- Jim Salzman. The $133 Million Bat Tunnel. Legal Planet.
- Louise Perry. Why does it cost £100m for HS2 to protect bats? BBC.



Izembek Controversy

- In a remote part of Alaska, two towns are separated by a wildlife refuge on an isthmus.
- Izembek National Wildlife Refuge is home to extraordinary biodiversity.
- People of King Cove want a road to Cold Bay through the refuge.
- This would improve quality of life for people of King Cove, but hurt wildlife.
- On his first day in office, Trump ordered expedited development of this road.
- The Izembek Refuge Road. Environmental and Energy Law Program - Harvard Law School. https://eelp.law.harvard.edu/tracker/the-izembek-refuge-road/



Discuss reading on roads and ag productivity

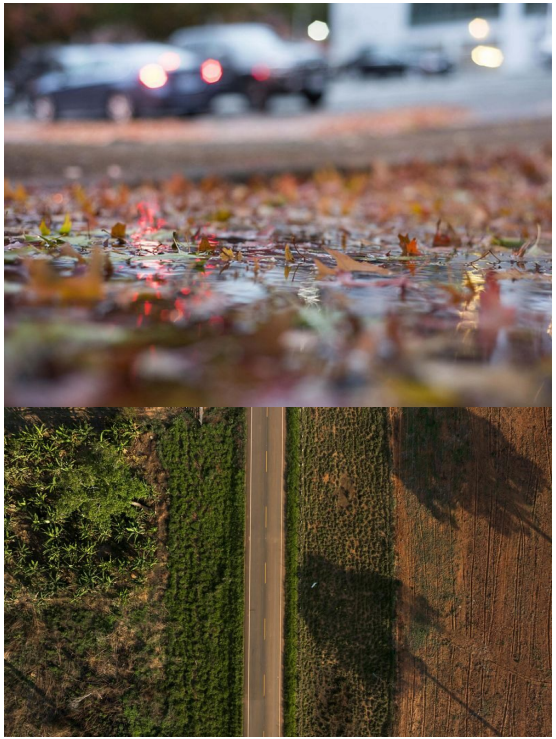
- Yogita Shamdasani's piece on "Rural road infrastructure & agricultural production."
- She wrote about the impacts of India's Pradhan Mantri Gram Sadak Yojana (Prime Minister's Village Road Program).
- Manasi can lead the discussion.



Pradhan Mantri Gram Sadak Yojana

Discuss reading on road impacts on fish

- Wagner wrote about 6PPD.
- What can be done about these impacts?
  - Not driving so much - VMT pricing
  - Driving gently
  - Bioswales
  - R&D into new tire technologies
- Johanson wrote about highway development in Brazil and Peru.
  - What can be done about this?
- Shirah can lead discussion.



Writing project

- Now you have done the op-ed, we can get started on the reports.
- I have developed a rubric which I have shared with you.
- Each remaining assignment will be graded by this rubric.

Criterion	Points
Grammar / punctuation	2
Scholarly technical writing style	3
At least 8 Sources	3
Factual accuracy	3
Meets word count	3
Considers nuances of issue - not just hammering a point	3
Makes a meaningful insights and contributions	3
Total points:	20

Writing Style - Technical vs Op-ed

Technical - specific in structure and format	Op-ed - much more open-ended
Avoids opinions and ‘should’ statements	Can include opinions, what you think should happen.
Avoids personal anecdotes	Can include personal stories and perspectives
Avoids rhetorical questions	Can include rhetorical questions as a hook
All statements must be cited and grounded in evidence.	Can include personal views.
Use more formal language	Can use colloquial language

Hypothesis Testing & Binomial Distribution

Week 4– 10/24– 10/26/23

TA: Narayan Gopinathan PP 203

Statistical Methods for Policy Analysis

Hypothesis Testing

Discuss readings

- In chapter 19 on hypothesis testing, DeVeaux made an analogy between hypothesis testing and juries.
- You can't accept a null hypothesis, but you can fail to reject it.
- You can't find a defendant innocent, but you can fail to prove guilt.
- That is why courts don't declare suspects innocent but "not guilty".

Hypothesis Testing

- A hypothesis is a falsifiable statement about a conceived relationships or existing condition in the world.
    - Falsifiable indicates it's possible that the statement could be disproven by data.
- A hypothesis test is a structured way to evaluate if an average differs from an expected amount or if a relationship exists between two or more variables.

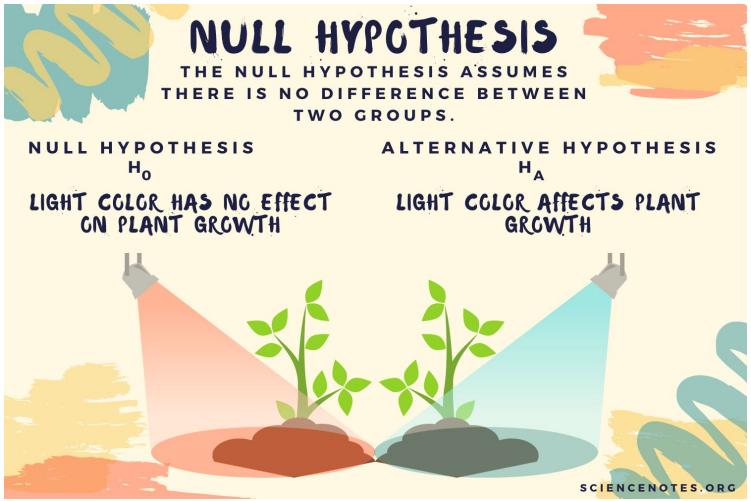
Six Steps of a Hypothesis Test

1. Create the null and alternative hypotheses.
2. Choose a statistical test.
3. Choose a critical value.
4. Calculate the test statistic.
5. Compare the test statistic to the critical value.
6. Make a conclusion about the state of the world.

Null & Alternative Hypotheses

Null hypothesis - the value against which you are measuring the outcome.

- It's usually 0, but doesn't have to be.
- Expressed in terms of μ because it's a **population parameter.**



Alternative hypothesis - what you the data will support.

- Two-tail hypothesis test = alternative hypothesis is "not equal to" (aka > or <)
  - Predicting there's a change, but not specifying in which direction that change occurs.
- One-tail hypothesis test = alternative hypothesis is "greater than" OR "less than," **but not both**.
  - Specifying that you predict an increase OR a decrease.

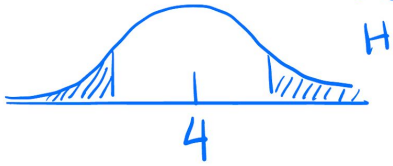
1- and 2-tailed Hypothesis tests

2-Tail Hypothesis Test

$H_0: \mu = 4$

$H_A: \mu \neq 4$

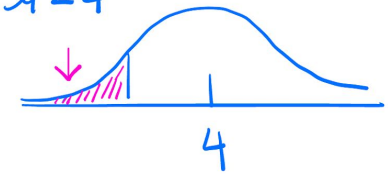
(Remember, μ doesn't have to be $\emptyset$)



1-Tail Hypothesis Test

$H_0: \mu = 4$

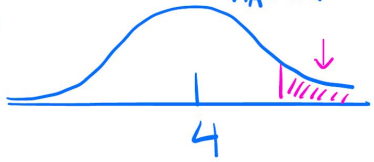
$H_A: \mu < 4$



OR

$H_0: \mu = 4$

$H_A: \mu > 4$



... but NOT both

"I predict a decrease in the average ... due to ..."

"I predict an increase in the average ... due to ..."

Critical Value (t*)

- The value to which you'll compare your test statistic to determine whether you reject or fail to reject Ho.
- Two tail hypothesis test:
  - Key words: "change", "does not equal"
- One tail: Only above or below
  - Key words: "greater than," "increases," "less than," "decreases."

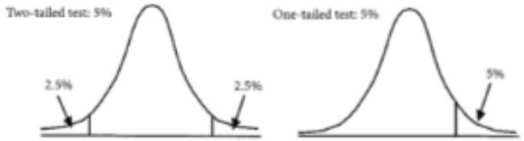
Choosing a Critical Value (t*)

T* depends on:

- Are we doing a 1 or 2 tailed hypothesis test?
- Alpha (derived from our confidence level)
- Degrees of freedom (sample size, n - 1)
- You may recall from last week, a higher level of confidence makes α smaller because α = 1 - Confidence Level (as a decimal)
    - Implication: The smaller α (the higher your confidence level), the harder it is to reject Ho. Do you understand this intuition?

Choosing a Critical Value (t*)

Appendix C
T-Test Distribution

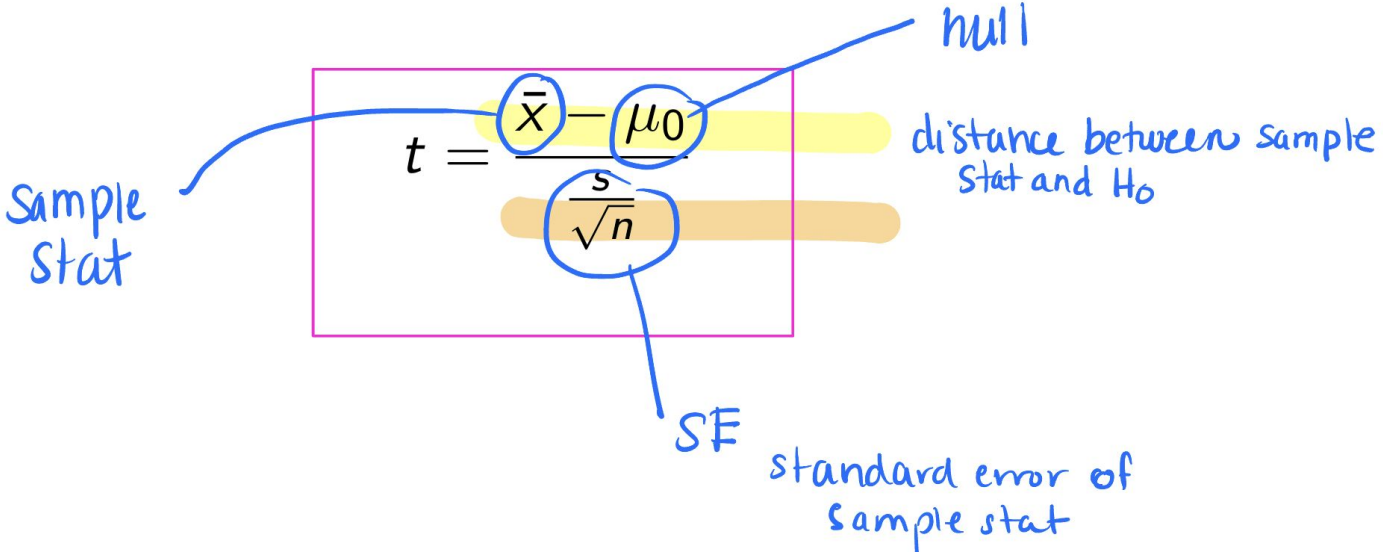


Degree of Freedom (df)	Alpha Level for One-Tailed Test: .10	.05	.025	.01	.005	.0025
	Alpha Level for Two-Tailed Test: .20	.10	.05	.02	.01	.005
1	3.078	6.314	12.706	31.821	63.657	127.32
2	1.886	2.920	4.303	6.965	9.925	14.089
3	1.638	2.353	3.182	4.541	5.841	7.453
4	1.533	2.132	2.776	3.747	4.604	5.598
5	1.476	2.015	2.571	3.365	4.032	4.773
6	1.440	1.943	2.447	3.143	3.707	4.317
7	1.415	1.895	2.365	2.998	3.499	4.029
8	1.397	1.869	2.306	2.896	3.355	3.833
9	1.383	1.833	2.262	2.821	3.250	3.690
10	1.372	1.812	2.228	2.764	3.169	3.581
11	1.363	1.796	2.201	2.718	3.106	3.497
12	1.356	1.782	2.179	2.681	3.055	3.428
13	1.350	1.771	2.160	2.650	3.012	3.372
14	1.345	1.761	2.145	2.624	2.977	3.326
15	1.341	1.753	2.131	2.602	2.947	3.286
16	1.337	1.746	2.120	2.583	2.921	3.252
17	1.333	1.740	2.110	2.567	2.898	3.222
18	1.330	1.734	2.101	2.552	2.878	3.197
19	1.328	1.729	2.093	2.539	2.861	3.174
20	1.325	1.725	2.086	2.528	2.845	3.153
21	1.323	1.721	2.080	2.518	2.831	3.135
22	1.321	1.717	2.074	2.508	2.819	3.119
23	1.319	1.714	2.069	2.500	2.807	3.104
24	1.318	1.711	2.064	2.492	2.797	3.091
26	1.315	1.706	2.056	2.479	2.779	3.067
27	1.314	1.703	2.052	2.473	2.771	3.057
28	1.313	1.701	2.048	2.467	2.763	3.047
29	1.311	1.699	2.045	2.462	2.756	3.038
30	1.310	1.697	2.042	2.457	2.750	3.030
40	1.303	1.684	2.021	2.423	2.704	2.971
60	1.296	1.671	2.000	2.390	2.660	2.915
120	1.289	1.658	1.980	2.358	2.617	2.860
∞	1.282	1.645	1.960	2.326	2.576	2.807

Source: Adapted from Table III of Ronald A. Fisher and Frank Yates, *Statistical Tables for Biological, Agricultural, and Medical Research*, 6th edition, Longman Group, Ltd., London, 1974. (Previously published by Oliver & Boyd, Ltd., Edinburgh). Used with permission of the authors and publishers.

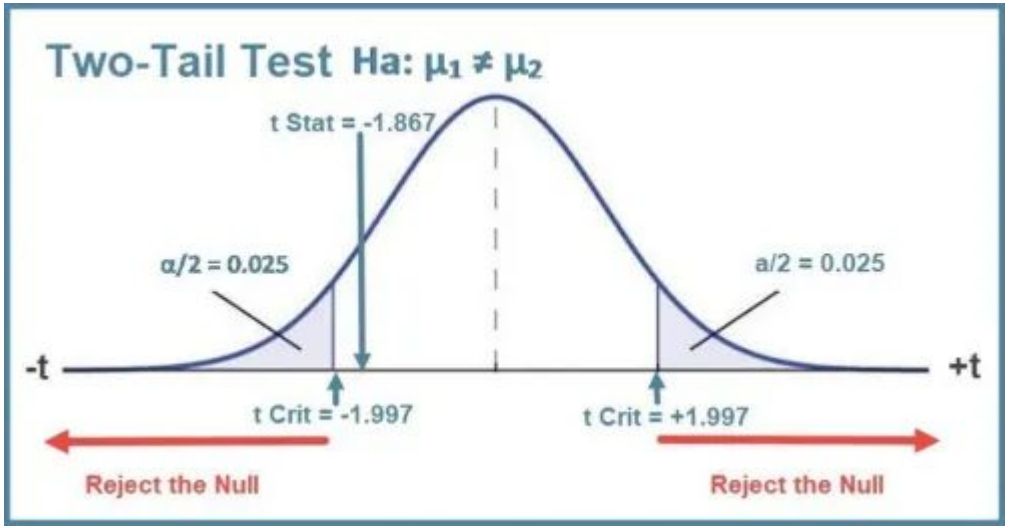
On exams, this will always be provided. Use this for the homework as well.

Calculating the Test Statistic



Comparing test statistic to t*

- Two tailed test: If |t| > |t*|, we reject the null hypothesis.
- One tailed test, looking for increase: If t > t* we reject the null hypothesis.
- One tailed test, looking for decrease: If t < - t*, we reject the null hypothesis.



Making a conclusion

- If you reject the null hypothesis, you can conclude with X% confidence that there is a **statistically significant** relationship between your two variables, or that there is a significant discrepancy between your observed and expected values.
    - This does not prove causality.
- If you can't reject the null hypothesis, then you can't conclude that. You can't state with confidence that there is a relationship between anything, but you can't state that there isn't.

Type I and Type II errors (not on exam/PS)

- Type I Error: The null hypothesis is true, but we reject it.
- Type II Error: The null hypothesis is false, but we don't reject it.

- In medical testing, the null hypothesis is that the patient is healthy.
- A type I error is a false positive, and a type II error is a false negative.

- In a jury trial, the null hypothesis is that the defendant is innocent.

- Which type of error is worse? Depends on the circumstances.

Relating Hypothesis Testing to Confidence Intervals (Week 3)

- Recall from last week, we calculate **confidence intervals** because when we take samples from a population, there's always variation. Thus, we **want to know if a sample falls beyond the normal variation we'd expect if Ho is true.**
- Hypothesis testing is part of the same idea. We want to know, "If I get this sample," say after I implemented a new program, "**how can I tell this value I observed for my sample is likely due to my new program** and not due to chance variation that would occur without my program?"
    - We say if the CI of the test stat does not overlap with my Ho, meaning the range of variation I could expect for my sample does not include my Ho, then it's very unlikely Ho is true. So, we reject it.

Question on hypothesis tests

- A governor is concerned about his negatives - the percent of voters who express disapproval of him. His political committee pays for a series of ads, hoping to keep the negatives below 30%. They then use follow-up polling to assess the effectiveness of the ad campaign.
- What are the null and alternative hypotheses that you would use for this scenario?
- Would you use a one or two tailed test to evaluate this hypothesis?

Question on hypothesis tests

- A governor is concerned about his negatives - the percent of voters who express disapproval of him. His political committee pays for a series of ads, hoping to keep the negatives below 30%.

- Null hypothesis: $p = 0.30$
- Alternative hypothesis: $p < 0.30$
- One-sided test.

Learning Objectives - Hypothesis Testing

How do we feel?

By the end of today's discussion, students will be able to

- Explain the purpose of hypothesis testing
- Set up and conduct a hypothesis test
- Interpret a hypothesis test

For Hypothesis Testing with Binomial Probability Model

- Distinguish between p and Pr(X=x) for a binomial probability model
- Identify when to reject or fail to reject a Ho

Binomial Model

What is it and how does this connect with what we've learned so far?

Learning Objectives - Binomial Probability Model

By the end of today's discussion, students will be able to

1. Explain the purpose of a binomial model
2. Recognize when to use a binomial model
3. Connect a binomial model to the previous concepts we've learned
4. Identify the three pieces of information needed for a binomial model
5. Calculate the probability of observing a given outcome in a binomial model
6. Calculate the mean and standard deviation of the expected outcome

Bernoulli trials

Bernoulli trials are repeated tests of a binary variable
They have three conditions

1. Trials are independent.
2. Probability of each option is fixed across each trial.
3. There are only two options:
    - true or false,
    - success or failure,
    - heads or tails,
    - yes or no.

Mean and standard deviation of Bernoulli trials

n = number of trials
p = probability of success of each trial
q= probability of failure of each trial = 1-p
→ p+q = 1
X = number of successes in n trials.
k = a given value for X

Mean of the distribution is the expected value.

$$P(X = k) = {}_nC_k\, p^k q^{n-k}$$

$${}_nC_k = \frac{n!}{k!(n-k)!}$$

$$Mean: \mu = np$$

$$SD \quad : \sigma = \sqrt{npq}$$

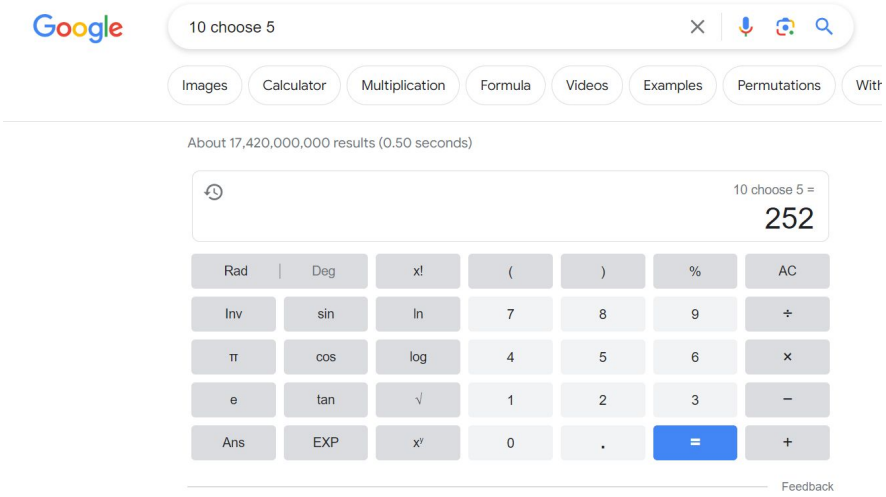
Factorial

- “n!” means n factorial.
- This is the number of ways you can arrange n items.
- This means the product of all integers between 1 and n.
- 1! is defined to be 1.
- 0! is defined to be 1.

11! = 1 * 2 * 3 * 4 * 5 * 6 * 7 * 8 * 9 * 10 * 11

Combinatorics

- ${}_nC_k$ is read as “n choose k”.
- N and k have to be nonnegative integers.
- $0 \leq k \leq n$
- Some calculators have this as a function
- Google can do it for you too.



$$C(n,k) = \binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

Binomial Probability Calculation (Broken Down)

Calculating probability of obtaining Given outcome

$$Pr(X = x \text{ or } k) = \underbrace{p^k \cdot (1-p)^{n-k}}_{\text{probability of } k \text{ successes in } n \text{ trials w/o order}} * \underbrace{\frac{n!}{k!\,(n-k)!}}_{\text{\# permutations}}$$

probability of success raised to # successes

probability of failure raised to # failures

(How many diff ways/orderings we can get k successes in n trials)

$$Pr(X = k) = [Eq\ 1] * [Eq\ 2]$$

or x

Six Steps of a (Binomial Probability) Hypothesis Test

1. Create the null and alternative hypotheses.
    a. Ho: **p = # (some "true" probability we obtained outside our study that we've been referencing)**
    b. Ha: **p (< or >) #**
2. Choose a statistical test.
    a. **Binomial model**
3. Choose a critical value.
4. **Calculate Pr(X (< OR >) k)**
5. Compare **Pr(X (< OR >) k)** to the critical value.
6. Make a grand, sweeping conclusion about the state of the world.

Example from lecture

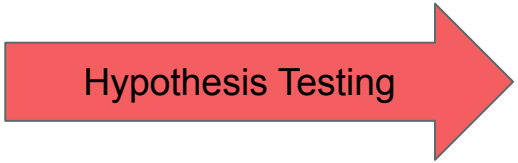
- In the example from lecture today (10/26), the professor used the example of getting 14 green lights out of 20.
- If we usually get 60% green lights (12 out of 20), that is our null hypothesis.
- Our alternative hypothesis is that something has changed - maybe LADOT has changed the algorithm, maybe there is a holiday, etc.
- To test the hypothesis, we add up the probabilities of getting 14 or more green lights if the null hypothesis is true.
- We calculate the chances of getting 14, 15, 16, 17, 18, 19, and 20 green lights, and add them up.
- If that sum is less than alpha, then, we reject the null hypothesis.

Inferential Stats

Big Idea: Story (Sample) → Big Picture (Population)

Binomial Probability Model of Our Sample

Components: n, p, k



Conclusion about Population
(the real-world outcome we care about)

So, when do we use the Binomial Probability Model?

In other words, what three assumptions must be true? This helps us also figure out the context clues in the problem.

Three Assumptions Are Needed

most useful for me to look for when reading a problem

1. Only two possible outcomes.
2. p is fixed across trials. The probability a fire becomes large is the same for each fire. The probability an arrestee is Black is the same for each arrestee. In other words, they give you a p (some universal probability not derived from sample data.
3. Trials are independent. The probability a fire becomes large is not affected by a previous fire. The probability an arrestee is Black is not affected by previous arrestees.

Practice Problem: Binomial Probability

Binomial Probability: First Gen Grad Students

Suppose that a sample of 7 Luskin students, 4 identify as first generation graduate students. According to the Council of Graduate Schools, 27% of doctoral students identify as first-generation. If that percentage is also accurate for master's students, then what is the probability that 4 of 7 Luskin students would be first generation?

What's our success (our outcome of interest)?

n =

k =

p =

Binomial Probability: First Gen Grad Students

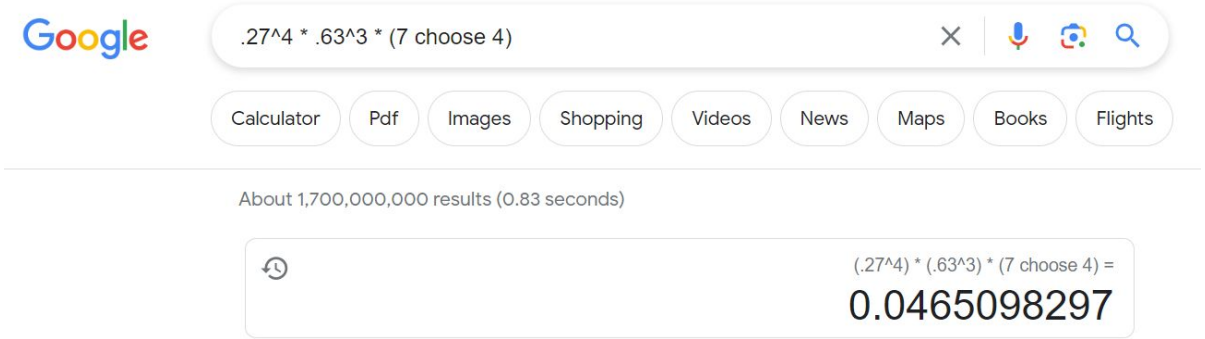
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What's our success (our outcome of interest)?

n = 7

k = 4

p = .27



Binomial Probability: Coin Tosses

- If you flip a coin 10 times, what is the probability of exactly 10 heads?
- Of exactly 5 heads?

Binomial Probability: Coin Tosses

- If you flip a coin 10 times, what is the probability of exactly 10 heads?
- Of exactly 5 heads?

- 10 heads:

$$.5^{10} \cdot .5^{0} \cdot \frac{10!}{10! * 0!} = .5^{10} = \frac{1}{1024} = .000976525$$

- 5 heads:

$$.5^{5} \cdot .5^{5} \cdot \frac{10!}{5! \cdot 5!} = .5^{10} \cdot \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{1}{1024} \cdot \frac{9 \cdot 8 \cdot 7 \cdot 6}{12} = \frac{252}{1024} = .24609375$$

Distinguished Teaching Assistant — Letter of Recommendation

This award celebrates Teaching Assistants who demonstrate a pattern of effective teaching that incorporates current best practices (e.g., student-centered learning, active learning, inclusive teaching), aims to inspire and challenge students, and supports the success of all students.

Name

Briefly describe how you know the nominee. (50 words max)

In a few sentences, describe a few ways in which the nominee inspired or challenged you or other students in the class. (200 words max)

Describe one or two things that the nominee did that you thought were particularly beneficial for your (or other students') learning or understanding of concepts. (200 words max)

Please share any additional reasons why you believe the nominee is an outstanding candidate for this award. (100 words max)

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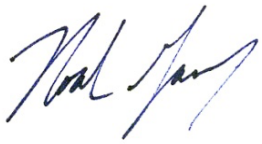
February 5, 2026

Dear Distinguished Teaching Award Committee:

I am writing in support of Narayan Gopinathan's nomination for the UCLA Distinguished Teaching Award for Teaching Assistants. I currently serve as the director for the Institute of the Environment's undergraduate Environmental Science Practicum capstone program, and teach a number of courses for IoES, including Introduction to Environmental Science, Foundations of Environmental Regulation and Policy, and Leadership in Water Management. I had the pleasure of having Narayan serve as my TA for the Leadership in Water Management course in the Fall of 2022. The fall 2022 quarter presented a number of extraordinary challenges for the UCLA campus as a whole, with a general strike by academic apprentices beginning part way through the quarter, coming on the heels of the global COVID-19 pandemic that had entirely halted in-person education. Despite they trying circumstances, Narayan established himself as an excellent and knowledgeable instructor, who was generous with his time to students and strongly empathetic to the challenges they faced both with course material and with the external pressures they were facing as a result of the pandemic. He was thoughtful in the ways he approached class discussions and above all, clearly conveyed his own passion for the material to the students.

Narayan's abilities as a Teaching Assistant were most clearly showcased, however, when academic apprentices and graduate researchers on campus went on strike in the latter half of the quarter, leaving faculty and lecturers scrambling to keep up with course and student needs, and students without a vital source of educational support. While supporting his fellow academic apprentices, Narayan also went out of his way to ensure that students in the Water Management class were not abandoned. He provided guidance to students as they worked towards completing a final policy and research project, showed up to the class's final presentation session to support the students in their work, and even helped a student navigate completing their required coursework when they came down with COVID-19 just before the final presentations of their quarter-long projects. Nartayan consistently demonstrated a dedication to his students and to his role as a teaching assistant in general, and I am happy to recommend him for this award.

Sincerely,



Noah Garrison
Environmental Science Practicum Director
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